AN EXPLICIT PRIME RANGE FOR THE HEIGHT-TWO COUNTEREXAMPLES TO THE HAHN–WILSON CONJECTURE

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Abstract. OpenAI's preprint *Counterexamples to the Hahn–Wilson conjecture at height two* constructs, for every sufficiently large prime p, a connective p-complete spectrum of exact fp-type two outside $\mathrm{Thick}(\mathrm{BP}\langle 2\rangle^{\wedge}_p)$. The paper states no numerical range. Its parameter conditions ($M \geq 81$, $p > 16M^2$, $2\lfloor M/8\rfloor > p^{1/3}$) first hold at $p = 17{,}073{,}437$. We show that, with one parameter chosen differently and the argument otherwise unchanged, the construction works for *every prime* $p \geq 293$, and that 293 is the least prime this argument can reach. A sharper reading of one step adds the primes 199 and 211. This note changes no idea of the upstream proof and assumes its correctness. It is a re-tuning of constants, recorded because the upstream statement gives no explicit bound.

1. Introduction

The Hahn–Wilson conjecture [LP26, Conjecture 1.1] asserts that a p-complete spectrum has fp-type at most n if and only if it lies in $\mathrm{Thick}(\mathrm{BP}\langle n\rangle^{\wedge}_p)$. The preprint [Ope26] refutes it at $n = 2$:

> *For every sufficiently large prime p there is a connective p-complete spectrum X of exact fp-type two with $X \notin \mathrm{Thick}(\mathrm{BP}\langle 2\rangle^{\wedge}_p)$, and $L^f_2 X \simeq L_2 X$, $L_{T(2)}X \simeq L_{K(2)}X$* [Ope26, Theorem 1.1].

The phrase "sufficiently large" enters through a single proposition, the construction of an operator with persistent defects [Ope26, Prop. `op:operator`]. All other sections need only $p > 8$ (algebraicity [Pst21, Cor. 6.9]) or $p > 5$. The operator is built under the parameter conditions

$$(\texttt{op:parameters}) \qquad M \text{ odd}, \qquad M \geq 81, \qquad p > 16M^2, \qquad 2\lfloor M/8\rfloor > p^{1/3}$$

together with $R_0 = \lfloor M/8\rfloor$. The upstream text remarks that "no particular numerical lower bound for p will be needed". The least prime satisfying (`op:parameters`) for some M is 17,073,437 (with $M = 1033$).

Theorem A. *Assume the arguments of [Ope26]. Then the conclusion of [Ope26, Theorem 1.1] holds for every prime $p \geq 293$. More precisely, Proposition `op:operator` of [Ope26] holds, with its proof unchanged except for the substitutions listed in Table 1, for every prime p such that some odd $M \geq 11$ satisfies*

$$(*) \qquad \frac{p+1}{2M} - \frac{p(M+1)}{p-1} > 1 \qquad \text{and} \qquad (M-3)^3 > p,$$

and these are exactly the primes $p \geq 293$.

Theorem B. *Within the same argument, with $R_0 = \lfloor M/8\rfloor$ replaced by a free R_0 but every other estimate as written, no prime below 293 can be reached. With the sharper reading of the cut argument in Remark 3.3, the reachable primes are exactly $\{199, 211\} \cup \{p \geq 293\}$.*

The improvement factor is $17{,}073{,}437/293 \approx 5.8 \times 10^4$. Removing only $M \geq 81$ and $p > 16M^2$ while keeping $R_0 = \lfloor M/8\rfloor$ gives 38,113. Most of the gain therefore comes from the choice of R_0.

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Scope. This note introduces no new idea. All mathematics is in [Ope26], and our statements are conditional on its correctness, which we have not independently verified beyond the numerical steps below. The bound 293 is limited by the correction estimate (2.1), which is the first estimate of the upstream proof to fail when p decreases. Going lower would require a different estimate for the coefficient-control lemma, not a different choice of parameters (Remark 3.4).

2. The numerical inputs of the operator construction

We use the notation of [Ope26, §op]: $q = p+1$, $\delta = (p-1)/2$, centers $d_i = \lfloor (i+\frac{1}{2})p/M \rfloor$ $(0 \le i < M)$, $s_0 = (M-1)/2$, $I = \{s_0 - R_0 + 1, \dots, s_0 + R_0\}$, $B_0 = 2R_0$, $\beta = \log_p B_0$, and $\kappa \le p/(p-1)$ from [Ope26, (op:kappa)]. Line numbers refer to `operator.tex` (in `build/sections/`) at the upstream commit cited in the bibliography. We list every place where p, M or R_0 enters an inequality. A full reading of all nine source files (`verification/`) found no other use of the size of p besides $p > 8$ and $p > 5$.

(C1) *Correction bound* (l. 322–327). For a depth $b \ge X/(2M)$ and $m < (M+1)X$ one needs $qb - r'_m > X$. Since $r'_m \le \kappa m$, it suffices that

$$\frac{p+1}{2M} - \frac{p(M+1)}{p-1} > 1. \tag{2.1}$$

(C2) *Exponent* (l. 503). $\beta > 1/3$, i.e. $(2R_0)^3 > p$. It is used in the nonmembership theorem of [Ope26] (`obs:nonmembership`) against an $O(N^{1/3})$ term whose exponent comes from $N_{p^s} \ge p^{3(s-1)}$ and does not depend on p.

(C3) *Interior central set* (l. 507). I meets neither 0 nor $M-1$, i.e. $1 \le R_0 \le (M-3)/2$.

(C4) *Degree* (l. 346 and the pivot rule). $M < p$.

(C5) *Same-cluster separation* (l. 590–599). Lower portions of two labels in one cluster differ by at most DX/p^2, which must be $< X/q$. Upstream uses $D = p/4 + 3$, valid for $R_0 = \lfloor M/8 \rfloor$. Since $I \cup (I-1)$ has diameter $2R_0$ and the center gaps are $p/M \pm 1$, the same reasoning gives $D = 2R_0 p/M + 3$. The condition is $D(p+1) < p^2$.

(C6) *First band* (l. 601–606). The $k_1 = 1$ band $X(1/M \pm 2/p)$ must satisfy: deviation $1/p + D/p^2 \le 2/p$; separation $1/M + 2/p < 2/M - 2/p$ from the $k_1 = 2$ band; lower end $X(1/M - 2/p) \ge X/(2M)$. These amount to $D \le p$ and $p \ge 4M$, with strict inequality for separation.

(C7) *Base stage* (l. 546–562). $d = \lfloor p/M \rfloor$ satisfies $qd < p^2$, $d \ge p/(2M)$ and $d \le \delta$. The centers are distinct and $d_{s_0} = \delta$.

(C8) *Cut argument* (l. 680–697). For large X a crossing pair has cluster index i with $|i - s_0| \le 1 + 3M/p + o(1)$. The text rounds this to "within three places", valid since $M < p$, and then needs $\{s_0 - 3, \dots, s_0 + 3\} \subset I$, i.e. $R_0 \ge 4$. The counting argument (l. 700–721) needs a center digit $< \delta$ in $d(I)$ and one $> \delta$ in $d(I-1)$, i.e. $R_0 \ge 2$.

(C0) *Global.* $p > 8$ (comparison and transfer sections), $p > 5$ (geometry and completion sections).

Lemma 2.1. *Let $M \ge 5$ be odd, $1 \le R_0 \le (M-3)/2$, and $p > 4M$ prime. Then* (C0), (C3)–(C7) *hold.*

Proof. (C0), (C3), (C4) are immediate. For (C5) and (C6), $2R_0 \le M-3$ gives $D \le p - 3p/M + 3 \le p$. Then $D(p+1) \le p^2 + 4p + 3 - 3p(p+1)/M < p^2$, because $M < p/4 < 3p(p+1)/(4p+3)$. Separation and the lower end are equivalent to $p > 4M$ and $p \ge 4M$. For (C7), $d \ge 4$ and $d \ge p/M - 1 \ge p/(2M)$ since $p \ge 2M$. Next $qd \le (p+1)p/M < p^2$, and $d \le p/5 < \delta$. Consecutive centers differ by at least $\lfloor p/M \rfloor \ge 1$, and $d_{s_0} = \lfloor p/2 \rfloor = \delta$. □

Proposition 2.2. *Proposition `op:operator` of [Ope26], with the substitutions of Table 1, holds at a prime p whenever there is an odd M with $R_0 = (M-3)/2 \ge 4$ satisfying $(*)$.*

Proof. Condition (2.1) gives $(p+1)/(2M) > M+2$, hence $p > 2M(M+2)-1 > 4M$. Lemma 2.1 gives (C0) and (C3)–(C7). (C1) is (2.1); (C2) is $(M-3)^3 > p$; (C8) is $R_0 \geq 4$. In the proof of [Ope26, Prop. `op:operator`], (`op:parameters`) is used only through (C1)–(C8). The coefficient-control lemma is used only at depths $b \geq X/(2M)$, which (C6) and (C7) guarantee. The upstream argument therefore goes through verbatim. □

3. Which primes satisfy the criterion

Write $f(p,M) = \frac{p+1}{2M} - \frac{p(M+1)}{p-1}$. For fixed M it is increasing in p, and for fixed p it is decreasing in M.

Lemma 3.1. *Every prime $293 \leq p < 1000$ satisfies $(*)$.*

Proof. For $293 \leq p < 512$ take $M = 11$. Then $8^3 = 512 > p$, and $f(p,11) \geq f(293,11) = 1 + \frac{259}{803} > 1$. For $397 \leq p < 1000$ take $M = 13$. Then $10^3 = 1000 > p$, and $f(397,13) = 1.272\ldots > 1$. □

Lemma 3.2. *Every prime $p \geq 1000$ satisfies $(*)$.*

Proof. Let $t = p^{1/3} \geq 10$, let e be the least even integer with $e^3 > p$, and put $M = e+3$. Then $e \leq t+2$, so $13 \leq M \leq t+5 \leq \frac{3}{2}t$, and $R_0 = e/2 \geq 5$. Moreover

$$f(p,M) \;\geq\; \frac{t^3}{3t} - \frac{1000}{999}(t+6) \;\geq\; \frac{t^2}{3} - 1.001\,t - 6.01,$$

which is > 1 at $t = 10$ and increasing for $t \geq 10$. □

Proof of Theorem A. Combine Proposition 2.2 with Lemmas 3.1 and 3.2. That no prime below 293 satisfies $(*)$ is part of Theorem B. The remainder of the proof of [Ope26, Theorem 1.1] (§`consequences`) uses only Proposition `op:operator` and $p > 8$. □

Proof of Theorem B. With every estimate as written, (C8) forces $R_0 \geq 4$, so (C3) forces $M \geq 11$. If $M \geq 11$ and $p \leq 283$, then $f(p,M) \leq f(283,11) = 0.866\ldots < 1$, so (C1) fails. The sharpened variant is treated in Remark 3.3. □

Remark 3.3 (Sharper cut argument). In (C8) the bound $|i - s_0| \leq 1 + 3M/p + o(1)$ is < 2 whenever $3M < p$. By (C6) this holds, so $|i - s_0| \leq 1$, and $R_0 \geq 2$ suffices for both uses in (C8). The criterion becomes: some odd $M \geq 7$ satisfies $(*)$. For $M = 7$, (2.1) needs $p > 2M(M+2) - 1 = 125$, which is incompatible with $4^3 > p$. For $M = 9$ one needs $199 \leq p < 216$: indeed $f(197,9) = 0.949\ldots$ and $f(199,9) = 1.060\ldots$. For $M = 11$ one needs $p \geq 293$ as before. The reachable set is therefore $\{199, 211\} \cup \{p \geq 293\}$, which is not upward closed: the fourteen primes $223, \ldots, 283$ are not reached.

Remark 3.4 (Beyond the correction estimate). Estimate (2.1) uses the crude depth bound $b \geq X/(2M)$. With the exact first-band depth $b \geq X(1/M - 2/p)$, the review scripts in `verification/scripts/` find that the condition first holds at 157 (with $R_0 \geq 4$) and at 109 (with $R_0 \geq 2$). Using this requires restating the hypothesis of [Ope26, Lemma `op:coefficient-control`] and the base-stage step (l. 312 and l. 551). We have not written this out, and we make no claim about these primes.

4. Changes to the upstream text

5. Computations and verification

`computations/pmin.py` checks (C0)–(C8) in exact rational arithmetic for each prime and each pair (M, R_0), without using Lemma 2.1. It confirms:

- the minimum 293 with witness $(M, R_0) = (11, 4)$;
- that no prime below 293 admits any (M, R_0);
- every prime in $[293, 1000)$, plus spot checks up to 10^{12};
- the variant set $\{199, 211\} \cup [293, 1000)$;

Location (`operator.tex`)	Change
l. 265–268 (`op:parameters`)	Replace by: M odd, $4 \le R_0 \le (M-3)/2$, (2.1) and $(2R_0)^3 > p$.
l. 269–273	Replace "no particular numerical lower bound" by the range $p \ge 293$ (Theorem A).
l. 324	Cite (2.1) directly instead of deriving it from (`op:parameters`).
l. 500	$R_0 = \lfloor M/8 \rfloor \to R_0$ as in the new `op:parameters`.
l. 503	Justify $\beta > 1/3$ by $(2R_0)^3 > p$.
l. 592	$(p/4+3)X/p^2 \to (2R_0p/M+3)X/p^2$.
l. 601	"The bounds on p" $\to$ "The bound $p > 4M$ (a consequence of (2.1))".
l. 694	"since $M \ge 81$" $\to$ "since $R_0 \ge 4$".
`main.tex` l. 57, 115, 129, 197–198; `consequences.tex` l. 14, 172–173, 187	Optional: "sufficiently large prime" $\to$ "prime $p \ge 293$".

TABLE 1. Every change needed for Theorem A.

- the upstream value 17,073,437 as a positive control.

An independent reviewer (an AI agent that did not write this note) read all nine upstream source files line by line, reimplemented the conditions separately (`verification/scripts/`), and reached the same values. It also checked upward closure exactly up to $2 \cdot 10^5$ and in closed form up to $2 \cdot 10^7$. Its report is `verification/adversarial-review.md`.

REFERENCES

[LP26] David Jongwon Lee and Piotr Pstrągowski. "The monochromatic Hahn–Wilson conjecture". In: *Inventiones Mathematicae* (2026). Published online September 7, 2026. DOI: 10.1007/s00222-026-01448-y.

[Ope26] OpenAI. *Counterexamples to the Hahn–Wilson conjecture at height two*. OpenAI Math Release preprint, September 26, 2026. https://github.com/openai/math/tree/08fcd618213aa43b0fcaff4526a8a48d66d02e55/preprints/Counterexamples-to-the-Hahn-Wilson-conjecture-at-height-two-September-26-2026. 2026.

[Pst21] Piotr Pstrągowski. "Chromatic homotopy theory is algebraic when $p > n^2+n+1$". In: *Advances in Mathematics* 391 (2021), p. 107958. DOI: 10.1016/j.aim.2021.107958.